

TABLEAU

Module 6 - Data Visualization — Part I

Standard Charts — Building the Right Chart for the Right Data

Charts Covered in This Module

Topic 1	The Golden Rule of Data Visualization — good vs bad charts, which chart when
Topic 2	Bar & Column Charts — Simple, Grouped, Stacked, 100% Stacked
Topic 3	Line Charts — Single line, Multi-line, Dual Axis
Topic 4	Pie & Donut Charts — when to use them and the 5-slice rule
Topic 5	Treemap — showing part-of-whole with size and colour
Topic 6	Bullet Chart — comparing performance against a target/benchmark
Topic 7	Heat Map & Highlight Table — colour-based comparison matrix

Dataset: TTL Dataset • 20 Rows • Step-by-Step Instructions Throughout

Dataset — Thane Tennis League (TTL) 2026

We will use a cricket tournament dataset throughout all 7 topics in this module. It is inspired by IPL-style T20 cricket. Each row represents one team's performance in one match. We have 8 teams, multiple matches, and several performance columns — perfect for every chart type we will build.

Dataset: Thane Tennis League (TTL) Match Stats

#	Team	City	Match	Runs	Wickets	Overs	4's	6's	Win/ Loss	Target	Phase
1	Mumbai Lions	Mumbai	Match 1	187	6	20	18	9	Win	175	League
2	Delhi Eagles	Delhi	Match 1	175	8	20	14	6	Loss	187	League
3	Chennai Kings	Chennai	Match 2	203	4	20	22	11	Win	189	League
4	Bangalore Royals	Bangalore	Match 2	189	7	20	17	8	Loss	203	League
5	Kolkata Tigers	Kolkata	Match 3	165	9	20	12	5	Loss	171	League
6	Pune Warriors	Pune	Match 3	171	5	20	15	7	Win	165	League
7	Hyderabad Hawks	Hyderabad	Match 4	195	3	20	20	10	Win	182	League
8	Rajasthan Riders	Rajasthan	Match 4	182	7	20	16	7	Loss	195	League
9	Mumbai Lions	Mumbai	Match 5	210	2	20	24	13	Win	198	League
10	Chennai Kings	Chennai	Match 5	198	6	20	19	9	Loss	210	League
11	Delhi Eagles	Delhi	Match 6	178	5	20	16	8	Win	166	League
12	Kolkata Tigers	Kolkata	Match 6	166	8	20	13	6	Loss	178	League
13	Mumbai Lions	Mumbai	QF 1	221	3	20	25	14	Win	207	Playoff
14	Hyderabad Hawks	Hyderabad	QF 1	207	6	20	21	10	Loss	221	Playoff
15	Chennai Kings	Chennai	QF 2	215	4	20	23	12	Win	199	Playoff
16	Pune Warriors	Pune	QF 2	199	7	20	18	8	Loss	215	Playoff

#	Team	City	Match	Runs	Wickets	Overs	4's	6's	Win/ Loss	Target	Phase
17	Mumbai Lions	Mumbai	SF 1	230	2	20	26	15	Win	218	Playoff
18	Chennai Kings	Chennai	SF 1	218	5	20	22	11	Loss	230	Playoff
19	Mumbai Lions	Mumbai	Final	245	1	20	28	16	Win	232	Playoff
20	Delhi Eagles	Delhi	Final	232	4	20	24	13	Loss	245	Playoff

Understanding the Dataset

Each row = one team's batting innings in one match.

Teams: Mumbai Lions, Delhi Eagles, Chennai Kings, Bangalore Royals, Kolkata Tigers, Pune Warriors, Hyderabad Hawks, Rajasthan Riders.

Columns: Runs scored, Wickets lost, Overs faced (always 20 in T20), Fours hit, Sixes hit, Win or Loss result, Target they were chasing, Phase (League or Playoff).

Mumbai Lions won the tournament — they appear in QF, SF, and Final.

Topic 1 — The Golden Rule of Data Visualization

Before we build a single chart, we need to understand one core idea: a chart is not just a pretty picture. A chart exists to communicate ONE clear message to the person reading it. If the message is not clear in 5 seconds, the chart has failed — no matter how colourful it looks.

The Golden Rule

Every chart you build must answer ONE specific question clearly. Before you drag anything onto the canvas, ask yourself: What question am I answering? Who is reading this? What do I want them to do after seeing it?

1.1 The 3 Questions to Ask Before Building Any Chart

#	Question	Why It Matters
1	What am I comparing?	<i>Are you comparing values across categories? Tracking change over time? Showing a part of a whole? The answer decides the chart type.</i>
2	What is the data type?	<i>Is your data a number, a date, a category, or a text field? Different data types suit different charts.</i>
3	How many categories do I have?	<i>2 items → simple comparison. 5 items → bar chart works. 20+ items → bar chart gets messy, consider a treemap or filter.</i>

1.2 Which Chart for Which Question?

Your Question	Best Chart	Cricket Example
Which team scored the most runs?	Bar Chart	Runs by Team — horizontal bars
How did runs change match by match?	Line Chart	Runs per Match over time
What share of total sixes did each team hit?	Pie / Treemap	% of Sixes by Team
Did we meet the target score? By how much?	Bullet Chart	Runs vs Target per team
Which team × match combo had most fours?	Heat Map	Fours: Team vs Match
How do Runs vs Wickets relate to each other?	Scatter Plot	Runs vs Wickets (Module 7)
How do League vs Playoff runs compare by team?	Grouped Bar	Runs: League vs Playoff side by side

1.3 The 5 Most Common Beginner Mistakes

✖	Mistake	Fix
1	Using a pie chart with 10+ slices	Too many slices make it unreadable. Use a bar chart or treemap instead.
2	Using a line chart for categories	Line charts are for time or continuous data. Categories (teams, cities) should use bars.
3	Using 3D charts	3D charts distort proportions and look impressive but lie about the data. Never use 3D.
4	Too many colours	Use colour only to highlight a difference. Using 8 different colours just creates noise.
5	Y-axis not starting at zero	Truncating the y-axis makes small differences look huge. Always start bar charts at zero.



Topic 2 — Bar & Column Charts



Bar & Column Charts

The most used chart in the world — best for comparing values across categories

Bar charts and column charts are the same thing — just rotated. A Column chart has vertical bars (up-down). A Bar chart has horizontal bars (left-right). Use column when comparing a few items. Use horizontal bar when you have long category names or many items.

Cricket analogy

Which team scored the most runs in the tournament? You want to compare 8 teams side by side. This is exactly what a bar chart does — each team gets one bar, and the length of the bar shows their total runs. The tallest bar = highest scorer. Instant answer.

2.1 Simple Bar Chart — Total Runs by Team

The most basic bar chart. One dimension (Team) on one axis, one measure (Runs) on the other.

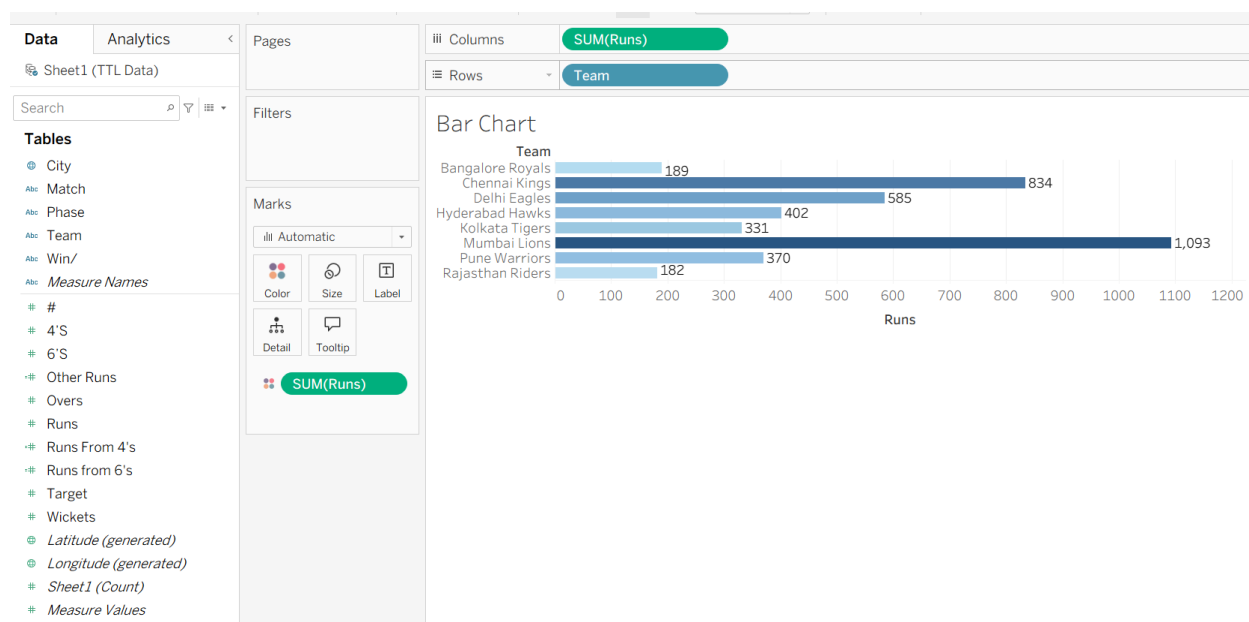
What the data looks like (aggregated for the chart):

Team	Total Runs	Matches Played	Chart Bar Length
Mumbai Lions	1,093	5 (including playoffs)	Longest — Tournament winners
Chennai Kings	834	4	Second longest
Delhi Eagles	585	3	Third
Hyderabad Hawks	402	2	Mid
Pune Warriors	370	2	Mid
Bangalore Royals	189	1	Short
Kolkata Tigers	331	2	Short
Rajasthan Riders	182	1	Shortest

Step-by-step — Build a Simple Bar Chart:

1	Connect Tableau to the cricket dataset (Excel or CSV)
2	Click on 'Sheet 1' at the bottom to open a blank worksheet
3	In the Data panel on the left, find the 'Team' field under Dimensions
4	Drag 'Team' to the Rows shelf (the horizontal bar at the top of the canvas)
5	Find 'Runs' under Measures in the Data panel

6	Drag 'Runs' to the Columns shelf — Tableau instantly draws a horizontal bar chart
7	To sort: click the Sort Descending button in the toolbar (bars go from longest to shortest)
8	To add labels on the bars: click the 'Label' button on the Marks card → check 'Show mark labels'
9	To change colour: click the 'Color' button on the Marks card → pick a colour you like
10	Give your chart a title: double-click the title at the top → type 'Total Runs by Team'



2.2 Grouped Bar Chart — Runs by Team and Phase

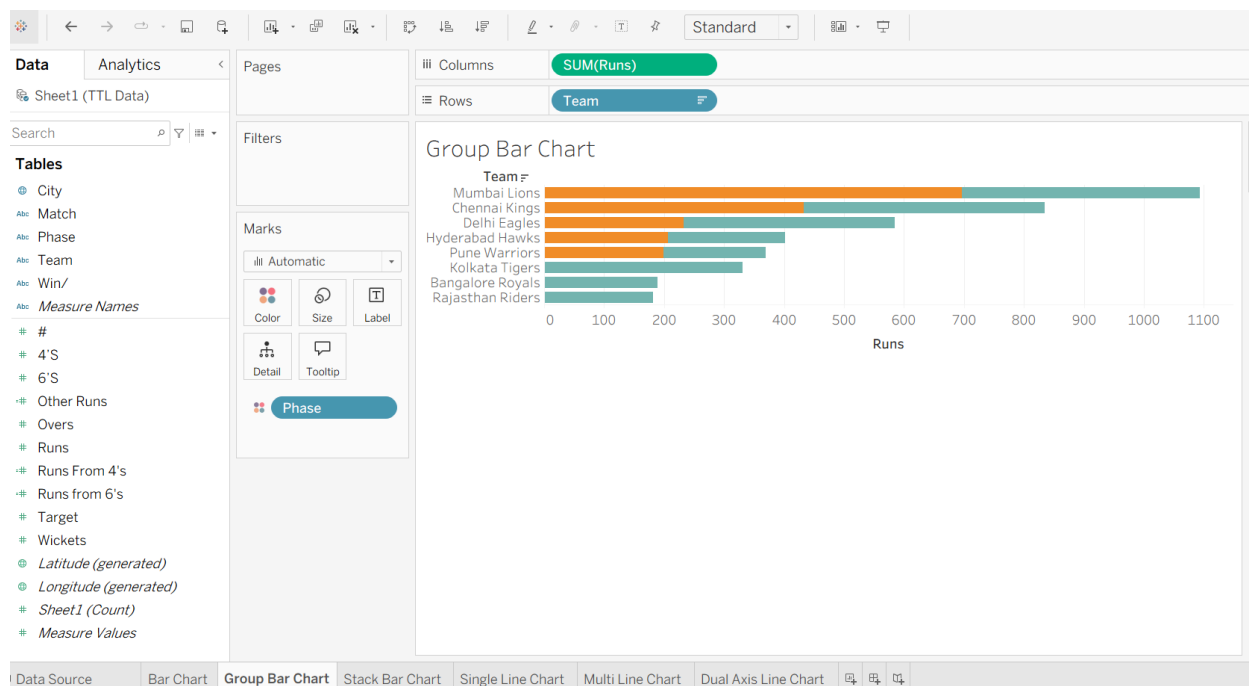
A Grouped Bar Chart shows multiple bars side by side for each category — one bar per sub-group. Here we compare League vs Playoff runs for each team.

When to use Grouped Bar?

Use it when you want to compare the SAME measure across two or more sub-categories. Example: League runs vs Playoff runs for each team. You see both numbers for each team simultaneously and can compare them directly.

1	Start with a blank worksheet
2	Drag 'Team' to the Rows shelf

3	Drag 'Runs' to the Columns shelf — you see one bar per team
4	Now drag 'Phase' (League / Playoff) to the Color shelf on the Marks card
5	Tableau automatically splits each team's bar into two coloured segments side by side
6	The bars are now grouped — blue for League, orange for Playoff (or whatever colours Tableau picks)
7	Sort by total runs: right-click 'Team' pill on Rows → Sort → Field → Runs → Sum → Descending
8	Add a legend title: click the legend on the right → Edit Title → type 'Tournament Phase'



2.3 Stacked Bar Chart — Runs Breakdown (Fours + Sixes + Other)

A Stacked Bar Chart splits each bar into segments stacked on top of each other. It shows both the total AND the composition at the same time.

Cricket example

Total runs are made up of: runs from fours (4×fours), runs from sixes (6×sixes), and runs from singles/twos/threes (other runs). A stacked bar shows each team's total runs AND how those runs were scored — all in one chart.

First, create two calculated fields:

Calculated Field 1: Runs from Fours

Formula: `[Fours] * 4`

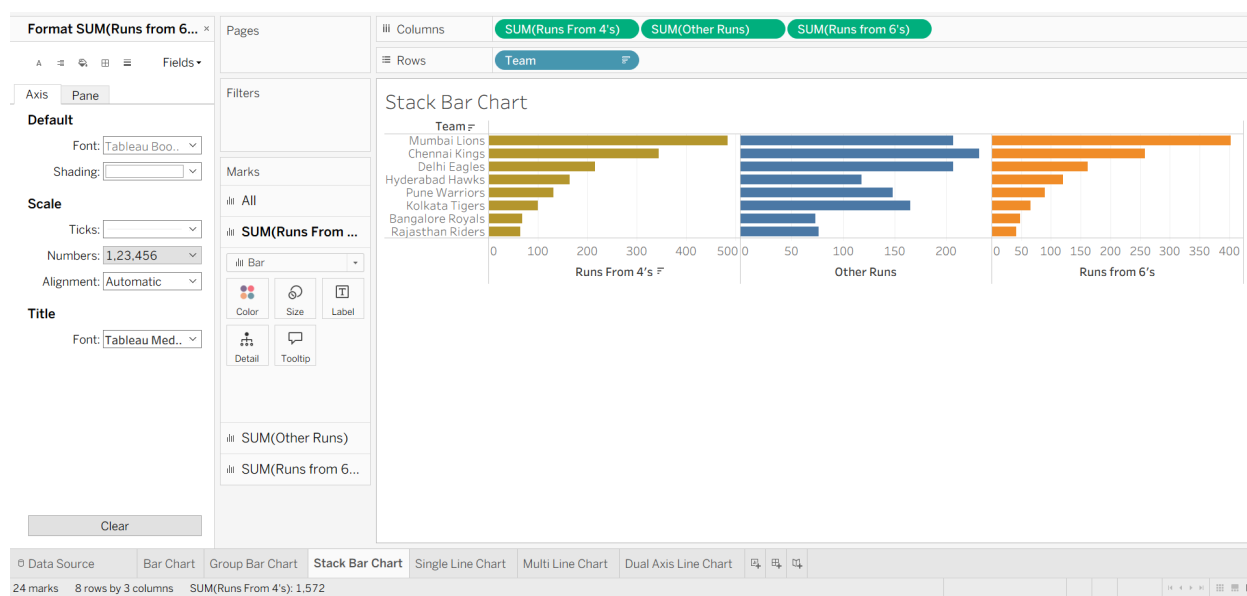
Calculated Field 2: Runs from Sixes

Formula: `[Sixes] * 6`

Calculated Field 3: Other Runs

Formula: `[Runs] - ([Fours]*4) - ([Sixes]*6)`

1	Create the 3 calculated fields above (Analysis → Create Calculated Field)
2	Drag 'Team' to the Rows shelf
3	Drag 'Runs from Fours' to the Columns shelf
4	Hold Ctrl and drag 'Runs from Sixes' to the Columns shelf — drops it next to Fours
5	Hold Ctrl and drag 'Other Runs' to the Columns shelf as well
6	You now have 3 separate mini bar charts — we need to stack them
7	Click on the 'Measure Names' pill on the Color shelf (Tableau puts it there automatically)
8	In the Marks card, change the chart type from 'Automatic' to 'Bar' using the dropdown
9	Now right-click the Columns shelf → select 'Stack Marks' if not already stacked
10	Each team's bar now shows 3 coloured segments: Fours runs, Sixes runs, and Other runs



2.4 100% Stacked Bar Chart — Proportion Comparison

A 100% Stacked Bar Chart makes every bar the same length (100%) and shows **ONLY** the proportions. You lose the absolute total but gain perfect proportion comparison across teams.

Regular Stacked Bar	100% Stacked Bar
Bar length shows TOTAL value	Every bar is same length (100%)
You can compare totals across teams	You CANNOT compare totals
Segments show absolute values	Segments show percentages only
Use when total matters	Use when proportion/share matters
Example: Mumbai bar is tallest (most runs)	Example: Which team relies most on sixes?

1	Build your Stacked Bar Chart following steps from section 2.3 above
2	Right-click on the x-axis (the Runs axis) → Edit Axis
3	In the Axis dialog, find 'Scale' section — check 'Stack Marks as percentage'

4

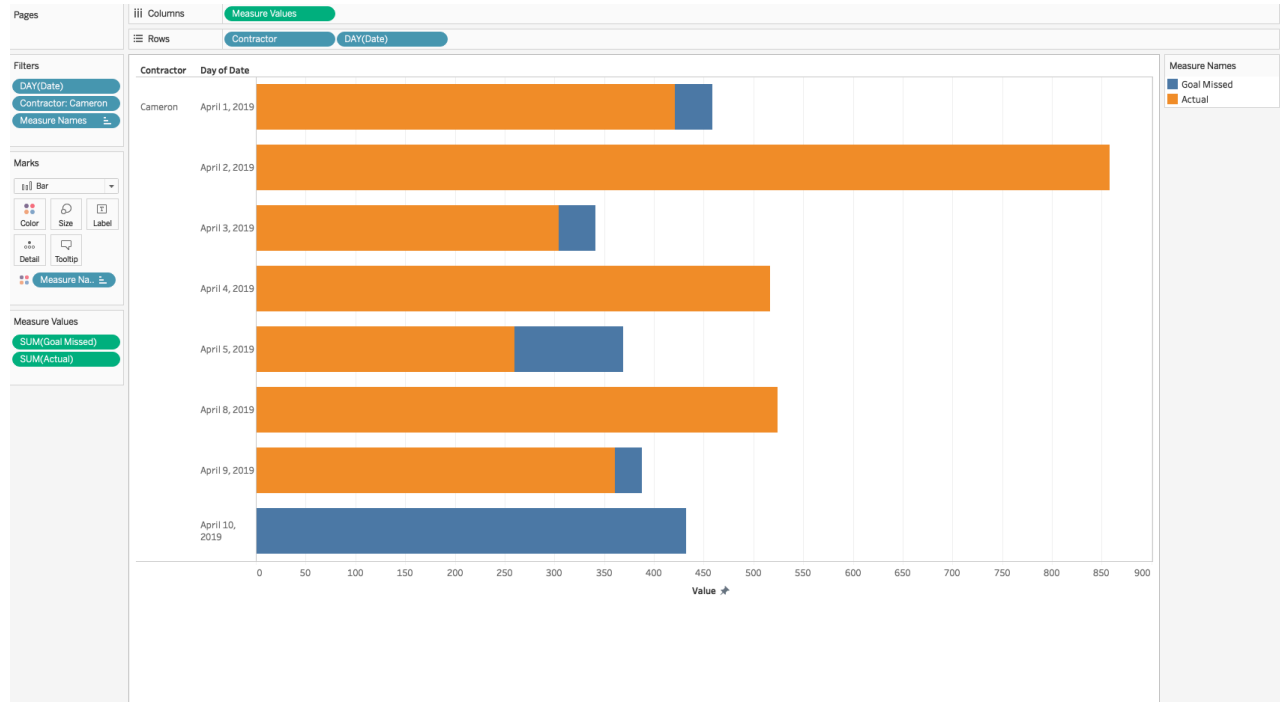
Alternatively: in the Analysis menu at the top → Stack Marks → choose 'On' and the axis converts to 100%

5

Now all bars are equal height and segments show % contribution

6

Add labels: click Label on the Marks card → show mark labels → you see % values on each segment



Topic 3 — Line Charts



Line Charts

Best chart for showing TRENDS over time — how something changes from point to point

A line chart connects data points with a line to show how a value changes over a sequence — usually over time. The direction of the line (going up, going down, flat) tells the story instantly. Use line charts for dates, matches in sequence, or any ordered data.

Cricket analogy

A team's run tally across matches is a trend. Did Mumbai Lions score more and more as the tournament progressed? Did their momentum build? A line chart shows this instantly — you see if the line is going up (improving), down (declining), or zig-zagging (inconsistent).

3.1 Single Line Chart — Mumbai Lions Runs Across Matches

Shows one team's run progression from Match 1 to the Final.

Match	Mumbai Lions Runs	Trend
Match 1	187	Starting point
Match 5	210	Improvement — +23 runs
QF 1	221	Growing stronger — +11
SF 1	230	Consistent growth — +9
Final	245	Peak performance — +15

1

In the Data panel, filter to show only Mumbai Lions (drag Team to Filters → select Mumbai Lions only)

2

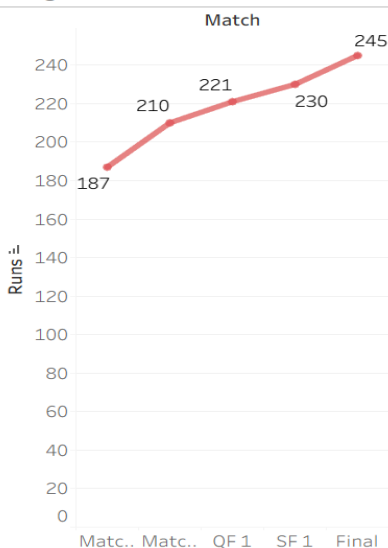
Drag 'Match' to the Columns shelf — you see match names appear as headers

3

Drag 'Runs' to the Rows shelf — Tableau draws dots at first

4	In the Marks card (left side of canvas), click the dropdown that says 'Automatic' → change it to 'Line'
5	A line now connects all the match points — you can see Mumbai's run trend clearly
6	Add data point markers: click the Color card → Markers → Show All — dots appear on each point
7	Add labels: click Label on the Marks card → Show mark labels → displays run values on each point
8	Title: 'Mumbai Lions — Run Progression Across Tournament'

Single Line Chart

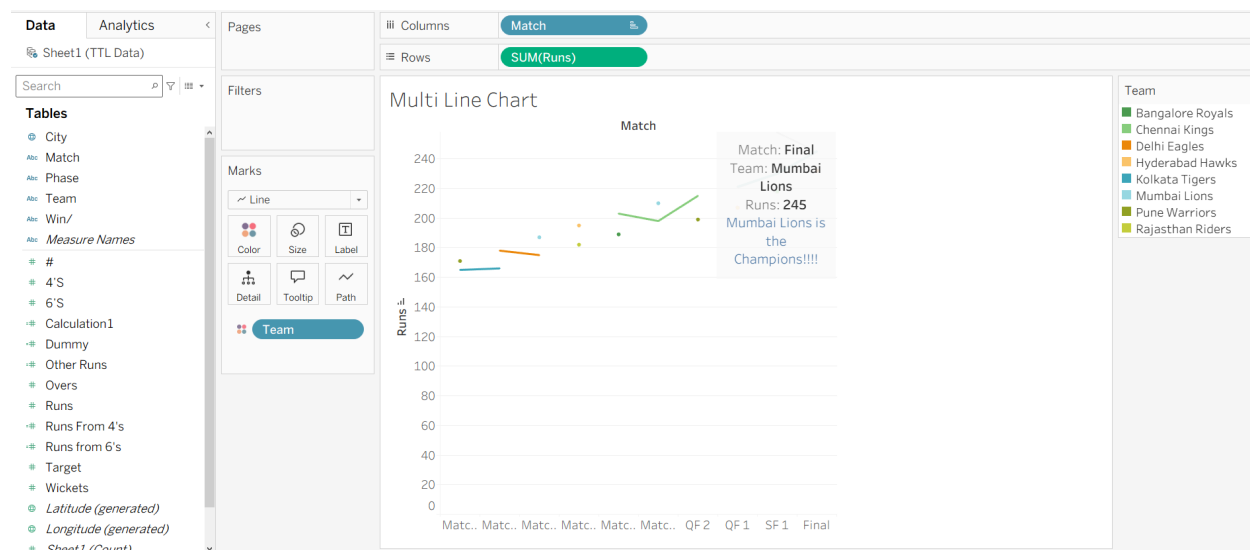


3.2 Multi-Line Chart — Comparing Multiple Teams

Shows multiple teams' run trends on the same chart using different coloured lines — one line per team.

1	Remove any team filters so all teams are visible
2	Drag 'Match' to the Columns shelf
3	Drag 'Runs' to the Rows shelf
4	Change the mark type to 'Line' in the Marks card dropdown

5	Drag 'Team' to the Color shelf on the Marks card — Tableau draws one coloured line per team automatically
6	A colour legend appears on the right side — each colour = one team
7	To highlight one team: click its name in the legend — other lines fade to grey
8	To add annotations: right-click any data point → Annotate → Mark — type a note (e.g. 'Mumbai wins the Final')



Tip — Too Many Lines?

If you have 8 teams all showing different coloured lines, the chart becomes a spaghetti mess. Use a filter to show only 2–4 teams at a time. Or highlight one team and dim the rest using the Set feature from Module 5.

3.3 Dual Axis Chart — Runs AND Wickets on One Chart

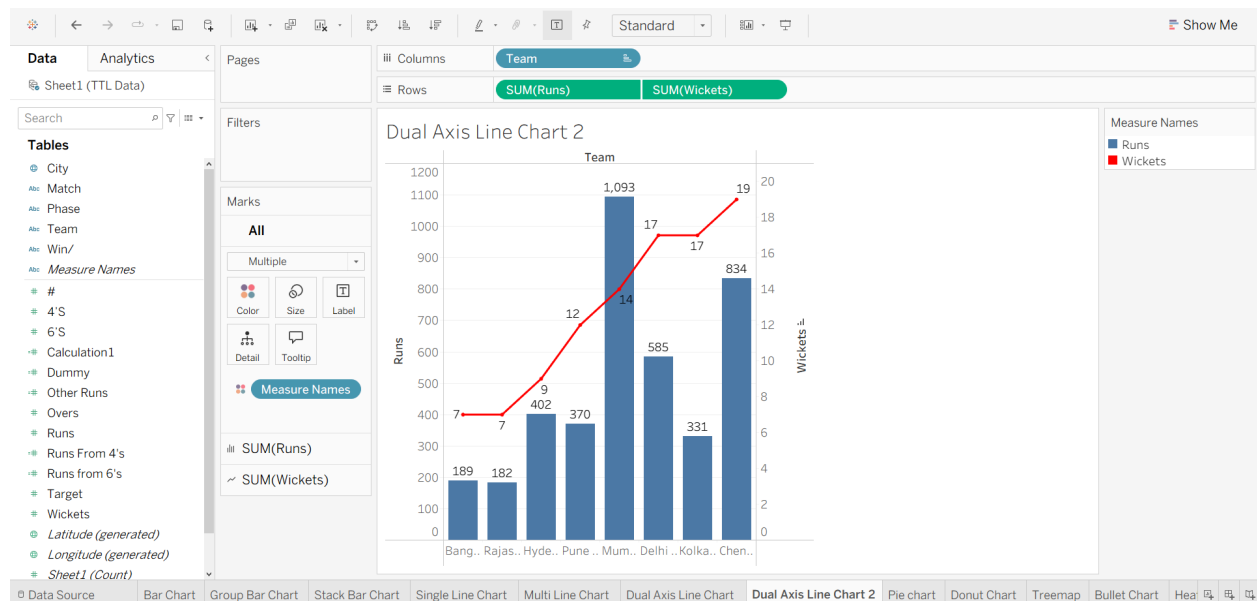
A Dual Axis chart combines TWO different measures on the same chart, each with its own axis on the left and right. This is very powerful — you can show Runs (left axis, bars) and Wickets lost (right axis, line) together.

Cricket analogy

A high score with few wickets lost means a dominant batting performance. A high score but 9 wickets lost means they struggled. Showing Runs (bars) and Wickets (line) together on one chart tells the full batting story.

1	Drag 'Team' to the Rows shelf
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2	Drag 'Runs' to the Columns shelf — you see bars
3	Now drag 'Wickets' ALSO to the Columns shelf — it appears as a second chart next to the first
4	Right-click the 'SUM(Wickets)' pill on the Columns shelf → click 'Dual Axis'
5	Wickets now shares the same chart space as Runs — they overlap
6	Right-click the right axis (Wickets axis) → Synchronize Axis — this aligns both axes so they don't mislead
7	In the Marks card, you now see two mark cards: one for Runs, one for Wickets
8	Change Runs to 'Bar' type and Wickets to 'Line' type — now you have bars for runs and a line for wickets
9	Change Wickets line to a different colour (e.g. red) so it stands out from the blue bars
10	Title: 'Runs Scored vs Wickets Lost by Team'



Topic 4 — Pie & Donut Charts



Pie & Donut Charts

Shows part-of-whole — what SHARE does each piece contribute to the total?

A Pie chart shows how a total is divided into parts — each slice is one category. The size of each slice shows its share of the total. A Donut chart is the same idea but with a hole in the centre — you can display a summary number in that hole.

Cricket analogy

8 teams played in the tournament. What % of total sixes in the entire tournament did each team hit? Each team gets a slice of the pie. The team that hit the most sixes gets the biggest slice. All slices together = 100% of all sixes.

4.1 The 5-Slice Rule — When Pie Charts Work and When They Fail

✓ Pie Chart Works Well	✗ Pie Chart Fails Badly
5 or fewer slices	6 or more slices — too crowded
Slices are clearly different in size	Slices are all similar size — impossible to compare
Showing simple part-of-whole	Comparing values across categories
Your audience asks 'what share?'	Your audience asks 'how much exactly?'
Example: Win % vs Loss % (2 slices)	Example: Runs by team (8 slices — use bar chart!)

The Hard Rule

Never use a pie chart with more than 5 slices. With 8 teams, a pie chart of runs by team is a disaster — 8 slices of similar size that no one can read. Use a bar chart instead. Save pie charts for simple 2–4 part comparisons only.

4.2 Build a Pie Chart — Win/Loss Split by Phase

Good use of pie: showing Win vs Loss split (only 2 slices!) for League phase matches.

1	Filter your data to League phase only: drag 'Phase' to the Filters shelf → select 'League'
2	In the Marks card dropdown, change chart type from 'Automatic' to 'Pie'
3	Drag 'Win/Loss' to the Color shelf — the pie splits into Win (one colour) and Loss (another colour)
4	Drag 'Number of Records' (or drag 'Match' and change to COUNT) to the Angle shelf — this controls slice size
5	Drag 'Win/Loss' also to the Label shelf — labels appear on each slice
6	Drag 'Number of Records' to the Label shelf as well — shows the count on each slice
7	To show percentage: right-click the measure on the Label shelf → Quick Table Calculation → Percent of Total
8	Resize the pie: make the chart window bigger or drag the pie to be larger
9	Title: 'Win vs Loss Split — League Phase'

4.3 Build a Donut Chart — Sixes Share by Team

A Donut chart = Pie chart with a hole. The hole can display a KPI number — like total sixes. This is a very popular style for dashboards.

How Tableau builds a Donut

Tableau does not have a native Donut chart button. We build it by layering TWO pie charts — one large (the real data) on top of a smaller white-filled circle that creates the hole effect. It uses a Dual Axis trick.

1	Create a calculated field: Donut Hole → type the number 0 (zero). This is a placeholder.
2	In Marks card, change chart type to 'Pie'
3	Drag 'SUM(Sixes)' to the Angle shelf — this sizes the pie slices by sixes count
4	Drag 'Team' to the Color shelf — each team gets a slice coloured differently

5	Now drag your 'Donut Hole' calculated field (value = 0) to the Rows shelf — a second chart appears
6	Right-click 'SUM(Donut Hole)' on the Rows shelf → Dual Axis
7	The two pie charts now overlap — one is the real chart, one is the hole
8	Click on the Donut Hole mark card → change its Color to White and its Size to roughly 60% (use the Size slider)
9	The white circle now sits in the centre creating the donut hole
10	Add a total label: drag 'SUM(Sixes)' to the Label shelf of the Donut Hole mark card → this shows total sixes in the centre

Topic 5 — Treemap

Treemap

Shows part-of-whole using RECTANGLES — size = value, colour = category or second measure

A Treemap shows data as nested rectangles. The bigger the rectangle, the bigger the value. It is great for comparing many items at once when a bar chart would be too long. It immediately shows you who dominates just by looking at the biggest box.

Cricket analogy

All 8 teams' total sixes side by side. The team that hit the most sixes gets the biggest rectangle. Mumbai Lions dominate with 67 sixes — their box would be much larger than Rajasthan Riders who only hit 7. You see the dominance instantly — no axis needed.

5.1 Treemap vs Pie Chart — When to Choose Which

Treemap	Pie Chart
Works well with 5–20+ categories	Only works with 5 or fewer categories
Rectangles easy to compare by size	Slices hard to compare when similar size
Shows two dimensions: size + colour	Shows one dimension: angle/size only
Good for showing dominance in a category	Good for simple part-of-whole with few items
Example: 8 teams × sixes — treemap wins	Example: Win vs Loss (2 items) — pie works

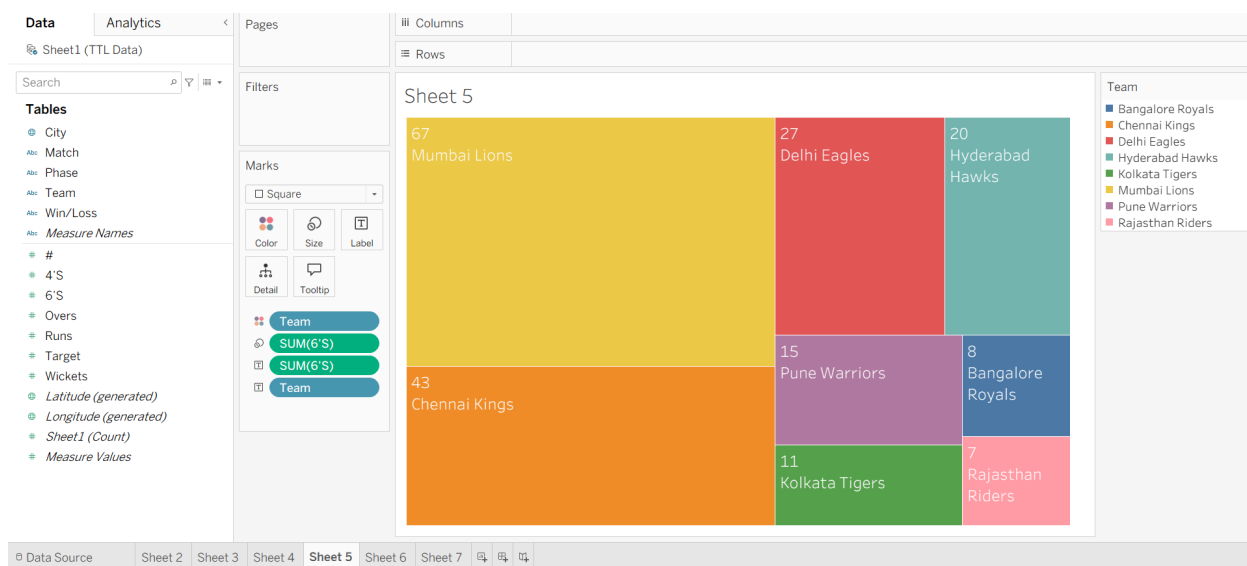
5.2 Build a Treemap — Sixes by Team

We will show how many sixes each team hit, with box size = sixes count and colour = team.

Team	Total Sixes	Relative Box Size
Mumbai Lions	67	Largest rectangle — tournament dominator
Chennai Kings	32	Second largest
Delhi Eagles	27	Third
Hyderabad Hawks	20	Mid-size
Pune Warriors	15	Mid-size

Team	Total Sixes	Relative Box Size
Kolkata Tigers	11	Small
Bangalore Royals	8	Small
Rajasthan Riders	7	Smallest rectangle

1	Open a new blank worksheet in Tableau
2	In the Marks card dropdown, change the chart type to 'Square' (this is Tableau's Treemap)
3	Drag 'Team' to the Label shelf on the Marks card — team names appear
4	Drag 'SUM(Sixes)' to the Size shelf on the Marks card — rectangles are now sized by sixes count
5	Drag 'Team' also to the Color shelf — each team gets a different colour rectangle
6	Drag 'SUM(Sixes)' also to the Label shelf — the sixes count appears inside each rectangle
7	Sort: right-click anywhere on the treemap → Sort → Descending by Sixes — biggest box goes to top-left
8	Resize the chart by dragging the chart window borders to make it larger and more readable
9	Title: 'Total Sixes by Team — Tournament Overview'



Topic 6 — Bullet Chart

Bullet Chart

Compare an ACTUAL value against a TARGET or benchmark — did we hit the goal?

A Bullet Chart is a special bar chart that shows two things together: the actual value (a bar) and the target value (a thin line crossing the bar). At a glance you can see: did the actual beat the target? By how much? It was invented specifically to replace dashboard gauges.

Cricket analogy

Every T20 innings, a team is chasing a target. Mumbai Lions in Match 1 scored 187 while chasing 175. Their bar = 187 runs. The target line = 175. The bar crosses the line — Win! In Match 2 if a team scored 180 chasing 195, the bar would stop before the line — Loss. This is a bullet chart in cricket form.

6.1 Anatomy of a Bullet Chart

Element	What It Represents	In Cricket Context
The main bar	The actual/achieved value	Runs scored by the team
The thin reference line	The target or benchmark	Target runs to win the match
Coloured bands (optional)	Performance zones: poor/ok/good	Below target / Near target / Above target
Bar longer than line	Actual beat the target	Team won the match
Bar shorter than line	Actual fell short of target	Team lost the match

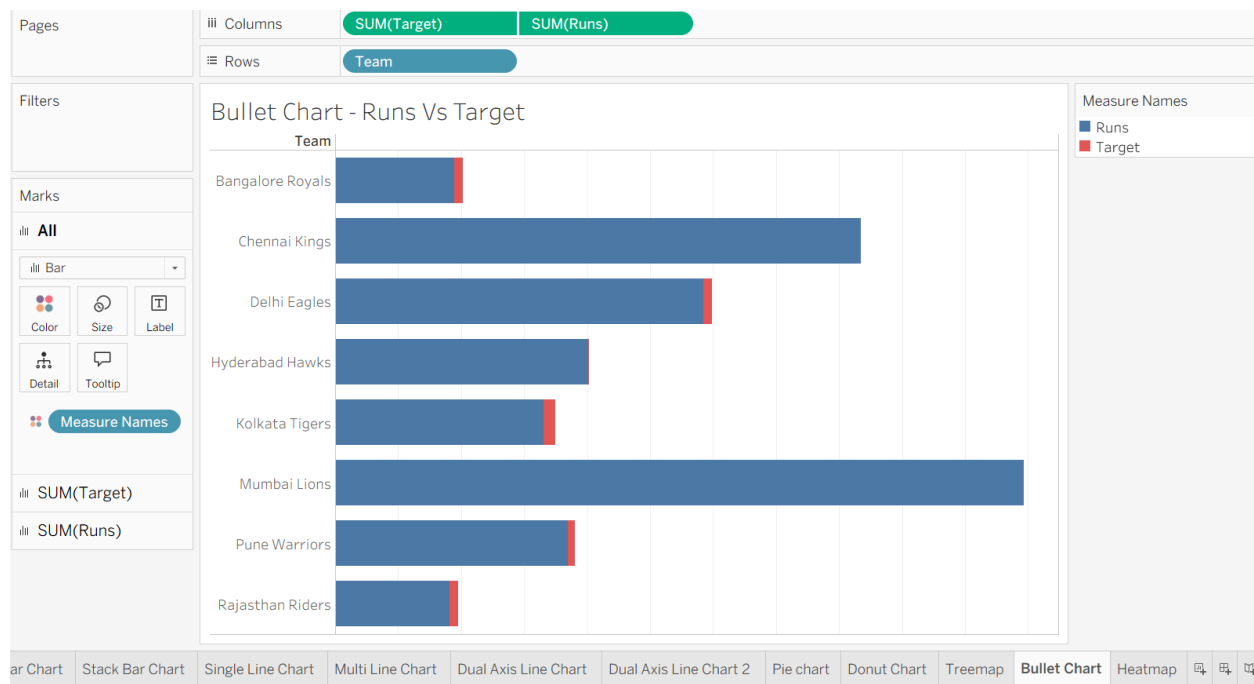
6.2 Build a Bullet Chart — Runs vs Target by Team

We will show each team's average runs scored vs their average target across all matches.

1	Open a new blank worksheet
2	Drag 'Team' to the Rows shelf
3	Drag 'Runs' to the Columns shelf — you see a bar for each team (average or sum, your choice)
4	Now drag 'Target' ALSO to the Columns shelf — a second chart appears next to the first

5	Right-click 'SUM(Target)' on the Columns shelf → click 'Dual Axis'
6	Target now shares the same axis as Runs
7	In the Marks card area, you see two mark cards: one for Runs, one for Target
8	Click the 'Target' mark card → change its mark type to 'Bar' — you now have two overlapping bars
9	Click the 'Runs' mark card → change its mark type to 'Circle' or 'Gantt Bar'
10	Click the 'Target' mark card → click Color → make it grey and reduce opacity to 70%
11	Now Target is the grey background bar and Runs is the coloured foreground — classic bullet chart look
12	Right-click the right axis → Synchronise Axis so both axes match
13	Right-click the right axis → uncheck 'Show Header' to hide the duplicate axis
14	Title: 'Runs Scored vs Target — Team Performance'

What the chart tells you:



Topic 7 — Heat Map & Highlight Table



Heat Map & Highlight Table

Compare many combinations at once — colour tells the story, no bars needed

A Heat Map uses colour intensity to represent a numeric value in a grid (matrix). Darker/brighter colour = higher value, lighter = lower. It is perfect when you have TWO dimensions and want to compare all combinations at once — like a 8×5 grid of teams vs matches.

Cricket analogy

You want to see every team's runs in every match in one single view. An 8-team × 5-match grid with colour: dark green = high runs, light = low runs. At one glance you see Mumbai's dark green in every match and Rajasthan's light colour. That is a Heat Map.

7.1 Heat Map vs Highlight Table — The Difference

Heat Map	Highlight Table
Shows colour only — no numbers visible	Shows numbers AND colour together
Colour intensity carries all meaning	Numbers in cells + colour as background
Better for large grids (10×10+)	Better for smaller grids where exact numbers matter
Use when trend/pattern is the story	Use when exact values + pattern both matter
Example: 8 teams × 10 matches colour grid	Example: 8 teams × 5 matches with run numbers shown

7.2 Build a Heat Map — Runs by Team and Match

We will create an 8-team × 6-match colour grid where darker green = more runs.

1	Open a new blank worksheet
2	Drag 'Team' to the Rows shelf — 8 team names appear as rows
3	Drag 'Match' to the Columns shelf — match names appear as columns
4	In the Marks card dropdown, change chart type to 'Square' (this creates the coloured grid squares)

5	Drag 'SUM(Runs)' to the Color shelf — Tableau automatically colours each cell by run count
6	Click the Color shelf → Edit Colors → choose a sequential palette (e.g. Green: light to dark)
7	Higher runs = darker green. Lower runs or empty matches = lighter or grey
8	Resize: go to Format → Cell Size → adjust to make squares bigger and more readable
9	Add borders: Format → Borders → Row Divider and Column Divider → set to a light grey line
10	Title: 'Runs by Team and Match — Heat Map'

7.3 Build a Highlight Table — Runs by Team and Match WITH Numbers

Same grid as the Heat Map but numbers are visible inside each coloured cell.

1	Follow all steps in 7.2 to build the Heat Map first
2	Drag 'SUM(Runs)' ALSO to the Label shelf on the Marks card (you already have it on Color)
3	The run numbers now appear inside each coloured cell — this converts the Heat Map to a Highlight Table
4	Format the numbers: right-click the label → Format → Number → choose 'Number (Standard)' with 0 decimal places
5	Adjust font: Format → Font → set label size to 9 or 10 so numbers fit inside the squares
6	Optional: drag 'Win/Loss' to the Shape shelf — a small icon appears in each cell showing W or L
7	Title: 'Team Performance Matrix — Runs by Match'

What the final Highlight Table looks like (snapshot):

Team	Match 1	Match 3	Match 5	QF 1	SF 1	Final
Mumbai Lions	187 	—	210 	221 	230 	245 
Delhi Eagles	175 	—	178 	—	—	232 
Chennai Kings	—	203 	198 	215 	218 	—
Hyderabad Hawks	—	195 	—	207 	—	—
Kolkata Tigers	—	165 	166 	—	—	—

 = Win (scored above target)  = Loss (scored below target) — = Did not play that match

Module 6 — Quick Reference Cheat Sheet

Chart Selection Guide

Question You Are Answering	Best Chart Type	Cricket Example
Compare values across categories	Horizontal Bar Chart	Total Runs by Team
Compare + show sub-group breakdown	Grouped Bar Chart	Runs: League vs Playoff by Team
Show composition of a total	Stacked Bar Chart	Runs = Fours + Sixes + Other
Compare proportions across groups	100% Stacked Bar	What % of runs came from sixes?
Show trend over time/sequence	Line Chart	Mumbai Lions runs Match 1 → Final
Compare 2+ trends on same chart	Multi-line Chart	All teams' run trends across matches
Show 2 different measures together	Dual Axis Chart	Runs (bars) + Wickets (line)
Show simple part-of-whole (≤5 items)	Pie Chart	Win vs Loss split in League phase
Show part-of-whole with centre KPI	Donut Chart	Sixes share by team + total in hole
Show many items by size comparison	Treemap	Sixes by all 8 teams
Show actual vs target performance	Bullet Chart	Runs scored vs Target to win
Compare many combos at once (pattern)	Heat Map	Runs by Team × Match (colour only)
Compare many combos (numbers+pattern)	Highlight Table	Runs by Team × Match (with numbers)

How to Build Each Chart — Quick Steps

BAR CHART:	Team → Rows		Runs → Columns		Sort descending
GROUPED BAR:	Team → Rows		Runs → Columns		Phase → Color

STACKED BAR: Values → Columns	Team → Rows Measure Names → Columns Measure
LINE CHART:	Match → Columns Runs → Rows Marks = Line
MULTI-LINE: Marks = Line	Match → Columns Runs → Rows Team → Color
DUAL AXIS: Wickets → Dual Axis	Runs → Columns Wickets → Columns Right-click
PIE CHART:	Marks = Pie Category → Color Measure → Angle
DONUT CHART: ~60%)	Pie chart + Dual Axis with white filled circle (size
TREEMAP:	Marks = Square Team → Label + Color Sixes → Size
BULLET CHART: background bar	Runs → Columns Target → Columns → Dual Axis grey
HEAT MAP: Runs → Color	Team → Rows Match → Columns Marks = Square
HIGHLIGHT TABLE:	Heat Map + Runs → Label shelf as well

The Golden Rules — Always Remember

Rule	Why It Matters
Ask 'What question am I answering?' first	Prevents building the wrong chart type
Never use pie with more than 5 slices	Too many slices = unreadable = bad communication
Line charts are for trends over time only	Using line for categories (teams) is wrong
Never use 3D charts	3D distorts proportions — visually misleading
Always start bar chart y-axis at zero	Truncated axis exaggerates small differences
Use colour to highlight, not decorate	Too many colours = noise, not information
Treemap beats Pie when you have 6+ items	Size comparison of rectangles is more accurate
Dual Axis: always Synchronize the axes	Unsynchronised axes tell a distorted story

End of Module 6 — Data Visualization Part I

Practice Exercise: Using the Cricket Tournament dataset, build all 7 chart types covered in this module.

Next → Module 7: Scatter Plots, Bubble Charts, Maps, Funnel, Dumbbell, Sankey & Word Cloud